

Price watch

PnS cameras, cabinets, portable hard drives and gaming keyboards get featured this month

**Mole in Microsoft?**

An ex-employee of Microsoft was arrested recently for leaking Windows 8 trade secrets. More here: <http://dgit.in/TraitorAlert>

A little too early for 4K

It may be all the rage at the moment, but now isn't the right time.

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The display segment is all the rage in 2014, we've got a lot of new terms being thrown around with 4K and UHD pretty much dominating the discussion. These two terms are used interchangeably but they really aren't the same.

Let's look into the whole high definition terminology. The current mainstream standard is 1920x1080 which is also termed as Full HD. Depending on the actual signal this may be 1080p or 1080i but let's put that one away for another day. The next step in display panel resolution is UHD or Ultra High Definition. This is achieved by doubling the horizontal and vertical pixel count of Full HD to get 3840x2160 resolution. 4K on the other hand stands for 4096x2160 which you can see is slightly greater than UHD. 4K operates on an aspect ratio of 1.9:1 while UHD has an aspect ratio of 1.78:1. Basically 4K is a production standard while UHD is a consumer format. If you were to view 4K content on a UHD panel then you'd see two

black vertical bars along either side of the screen. Manufacturers call UHD as 4K just because it sounds a bit snazzy and that's about it.

Pushing four times the pixels requires a heck lot more graphical processing power and current cards are already capable of the same. It's been quite some time since multi-display technologies have hit the market but the implementation is a bit different. There is a limitation on HDMI which prevents it from pushing UHD resolution on a single HDMI port. But multi-display is all about using each port for just one Full HD display.

Mobiles have started using beefier SoCs which can render UHD resolution, the Qualcomm Snapdragon 800 is one such chipset that has already made its presence felt in some flagships. If that wasn't enough then there are TV channels that have started projecting in 8K resolution.

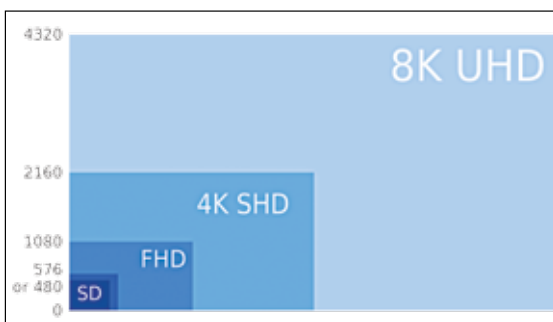
And by 8K I mean QUHD (Quad Ultra HD) which requires much higher bandwidth.

So with all the UHD capable devices around us and with the availability of content in UHD format, wouldn't it be time to move to a UHD monitor? Nope, and here's why. All 4K (that's what manufacturer's call UHD) monitors out there are using TN

(twisted nematic) panels. These only reproduce the image properly for the viewer sitting right in front of the monitor. Anyone sitting at an angle is going to experience a washed out display with a darkened overall tone. TN is great for fast response time but offers not so great colour reproduction. IPS and VA panels in UHD monitors are a while away.

If you've got any intention of gaming on a UHD monitor then you're going to be most disappointed. Firstly, the GPUs can only push UHD resolution at 30Hz over HDMI, this means that you'll be stuck with crappy FPS which would likely get capped at around 30 FPS given that a single GPU is going to find it ridiculously stressful to push UHD at max configuration. On top of that the sub 60Hz panel will worsen your overall experience. You can play at 1080p on your UHD monitor at 60Hz. However, you'd smack yourself in the head for buying the UHD monitor if you were never going to be able to play games at UHD resolution. Also Full HD looks really worse on an UHD monitor since each pixel gets stretched to cover 4 square pixels.

So the way manufacturers have gone about fixing this is by emulating one panel to seem like two. Hence, by using MST one display panel is made to seem like two on the DisplayPort. This allows UHD @60Hz on current "4K" panels, Dell's UP3214Q uses this method to get UHD onto an 32-inch IPS panel for a good \$3,500. So at the end of the day you either plunk down a ridiculous amount of money to get a proper UHD panel or you can spend close to \$800 and get TN panels running at 30Hz. I'd say give it a year or two, HDMI 2.0 which is capable of carrying UHD signals was released in September last year and it'll be some time before that becomes mainstream. That's when you should go in for a UHD monitor and a good graphics card too. **[1]**



4K and UHD aren't the same